

CS610 COMPUTER NETWORKS

Lecture 01: INTRODUCTION

1. What is a Network?

A network is a system that connects computers together using a single transmission technology so they can send and receive data from each other.

Hint: Network = computers ko aik dusre se connect karna

2. What is the Internet?

The Internet is a collection of many networks connected by routers, so that any computer on any network can communicate with any other computer.

Hint: Internet = networks ka network, routers se jura hua

3. How are Networks Classified? (4 ways)

i. By Size:

i.a. LAN (Local Area Network) — small area like a room, floor, or building

i.b. WAN (Wide Area Network) — large area like cities and countries; the Internet is a WAN

LAN vs WAN differ in speed, bandwidth, latency, security, cost, and standards

Hint: LAN = choti jagah (office/ghar), WAN = bara area (poora mulk)

ii. By Connectivity (Topologies)

ii.a **Point-to-Point:** Star topology (each computer connects to a central hub) and Tree topology

ii.b **Broadcast:** Satellite/Radio topology and Ring topology

Hint: Star mein hub hota hai center mein, Ring mein computers loop mein jure hote hain

iii. By Medium

iii.a Copper wire, Coaxial cable, Optical fiber, Radio waves

iii.b Each differs in speed, range, cost, and reliability

Hint: Optical fiber sabse fast hai, radio waves ki range sabse zyada

iv. By Mobility

iv.a Fixed networks vs Mobile networks

iv.b Mobile networks ke issues: location tracking, semi-persistent connections, complex billing

Hint: Mobile networks new hain, location tracking mushkil kaam hai

4. Why is Networking Complex?

a. Many different technologies exist (incompatible standards)

b. No single underlying theory

c. Models are either too simple or too complex

d. No uniform terminology

Hint: Complexity ki wajah = har company ka apna standard, koi common theory nahi

5. How to Master This Complexity?

a. Focus on concepts, not details

- b. Learn networking terminology
- c. Understand analogies, not just depth

Lecture 02: MOTIVATION AND TOOLS

1. Resource Sharing:

The main motivation for networking was resource sharing — making programs, equipment, and data available to anyone on the network regardless of physical location.
Example: One printer shared among all office workers instead of giving each person their own printer.

Hint: Resource sharing = printer, data sab ke liye available, location matter nahi

2. ARPA and ARPANET:

ARPA (Advanced Research Project Agency) started investing in data networking using packet switching and internetworking. This led to ARPANET — the first network, which eventually grew into the Internet in the 1970s.

Hint: ARPA ne research ki, ARPANET bana, phir internet bana — 1970s mein

3. Internet Growth:

The Internet has grown exponentially — doubling roughly every 9 to 12 months.

Hint: Log scale pe growth linear dikhti hai = exponential growth hai

4. Probing Tools:

4.1 **PING:** Sends packets to a destination to check if it is online. It reports round-trip time and packet loss.

4.1.1 **Problem:** If ping gets no response, it cannot tell WHY (could be remote computer issue, local issue, or congestion). Some computers reject ping to avoid flooding attacks.

4.2 **Trace Route:** A more detailed tool that shows every computer (hop) between source and destination, along with their response times.

Hint: PING = zinda hai ya nahi check karo, Trace route = poora rasta dikhao

Lecture 03: OVERVIEW OF DATA COMMUNICATION

This lecture is just an overview. Chapters 4, 5, and 6 are from the Data Communication course. They will NOT appear in assignments, quizzes, or exams. Only watch the video for context.

Lecture 04: PACKETS, FRAMES AND ERROR DETECTION

1. Why Packets?

Data is divided into small blocks called packets for two reasons:

- a. Sender and receiver need to coordinate to detect transmission errors

b. Individual direct connections between every pair of computers are not practical

Hint: Packets = data ke chote tukde, sharing easy hoti hai

2. **Problem with Sharing Without Packets:**

If a 5MB file is sent over a 56Kbps network without packets, it takes 12 minutes – all other computers must wait 12 minutes. This is unfair.

3. **Solution – Packets:**

Breaking the file into 1000-byte packets means each packet takes only 0.143 seconds. Other computers only wait 0.14 seconds before they can start transmitting.

Hint: Packets se fairness aati hai – sab computers turns le sakte hain

4. **TDM (Time Division Multiplexing):**

Packets allow TDM – each packet is sent through a multiplexer onto a shared channel, then a demultiplexer sends it to the correct destination.

Hint: TDM = shared channel pe turns mein data bhejte hain

5. **Packets vs Frames:**

a. Packet: Generic term for a small block of data

b. Frame (or Hardware Frame): A packet with a specific format for a specific hardware technology

Hint: Frame = packet ka specific format hardware ke liye

6. **Frame Format:**

Frames use a Header (SOH = Start of Header) and Tail (EOT = End of Tail) to mark boundaries of data.

7. **Framing Problems:**

a. Missing EOT means the sending computer crashed

b. Missing SOH means the receiving computer missed the beginning

c. Bad frames are discarded

Lecture 05: BYTE STUFFING

1. **The Problem (Data Stuffing):**

Sometimes special characters (SOH, EOT) appear inside actual data. The receiver would mistakenly think it is the start or end of a frame.

2. **Solution – Data Stuffing:**

The sender slightly modifies data before sending. Two types:

a. **Byte Stuffing** – for character-oriented hardware

b. **Bit Stuffing** – for bit-oriented hardware

3. **How Byte Stuffing Works:**

ESC (escape) character is inserted before reserved bytes.

For example:

- i. ESC + X replaces SOH
- ii. ESC + Y replaces EOT
- iii. ESC + Z replaces ESC itself

The receiver reverses these replacements.

Hint: Byte stuffing = data mein agar special character aaye toh ESC laga do taake confusion na ho

4. **Transmission Errors:**

Interference or power surges can flip bits. Error detection mechanisms are added to frames.

5. **Parity Checking:**

- a. **Even Parity:** Total number of 1-bits must be even
- b. **Odd Parity:** Total number of 1-bits must be odd
- c. A parity bit is added to make the count even or odd

6. **Limitation:**

Parity can only detect errors that change an odd number of bits. If two bits flip, parity misses it.

Hint: Parity = extra bit add karo, receiver count kare aur compare kare

7. **Checksum:**

Data is treated as integers, added up, and the sum is sent with data. Receiver recalculates and compares.

8. **CRC (Cyclic Redundancy Check):**

Data is treated as coefficients of a polynomial and divided by a known polynomial. The remainder is sent as CRC. Very reliable — detects both vertical errors and burst errors. Uses two hardware components: Shift Register and XOR unit.

Hint: CRC sabse reliable hai, hardware mein implement hota hai

Lecture 06: SHIFT OPERATION

1. **CRC Hardware:**

A 16-bit CRC uses three shift registers and three XOR units. Registers are initialized to zero, bits are shifted through, and after all bits are processed, the CRC value sits in the registers.

2. **Error Types CRC Detects Better:**

Vertical Errors: Hardware failure causes the same bit position in every character to be damaged

Burst Errors: A set of bits near a specific location change together (e.g., lightning strike)

Hint: Vertical = ek column damage, Burst = continuous bits ek jagah damage

3. **Frame Format with CRC:**

The final frame includes: SOH | Data | EOT | CRC

4. **Direct Point-to-Point Communication (Old Method):**

Each computer had a direct cable to every other computer. This forms a mesh network.

5. **Advantages:**

Different connection types, easy security

6. **Disadvantages:**

a. Connections grow as $n(n-1)/2$ for n computers — very expensive

b. Adding one computer needs $N-1$ new connections

Hint: Direct connections = bahut expensive, formula $n(n-1)/2$ yaad rakho

Lecture 07: GROWTH OF LAN TECHNOLOGY

1. **Why LAN?**

LAN reduces the number of connections by making all computers share a single cable/medium. Computers take turns.

Hint: LAN = ek cable sab share karte hain, cheap and simple

2. **Locality of Reference Principle:**

a. **Spatial Locality:** Computers tend to communicate with nearby computers

b. **Temporal Locality:** Computers tend to communicate with the same computers repeatedly

Hint: Aas paas waale aur zyada baat karte hain — isliye LAN ka concept useful hai

3. **Three Main LAN Topologies:**

a. **Star:** All computers connect to a central hub. A cable cut to one computer only affects that one computer.

b. **Ring:** Computers connected in a closed loop, passing data one-by-one. A cable cut disables the whole network.

c. **Bus:** All computers on a single long cable. A cable cut disables the whole network, but fewer wires needed.

Hint: Star = safe (ek cut = ek computer down), Ring/Bus = ek cut = sab down

4. **Ethernet:**

Invented at Xerox PARC in the 1970s. Uses bus topology with a single coaxial cable. Standards managed by IEEE (802.3).

a. Original speed: 3 Mbps

b. Current standard: 10 Mbps

c. Fast Ethernet: 100 Mbps

d. Gigabit Ethernet also available

Uses Manchester Encoding (signal changes to encode data).

Hint: Ethernet = sabse common LAN, IEEE 802.3 standard, bus topology

Lecture 08: CSMA (Carrier Sense Multiple Access)

1. **CSMA:**

Ethernet uses CSMA to coordinate who transmits when. A computer listens (senses the carrier) first:

a. If the cable is idle, it transmits

b. If the cable is busy, it waits

Hint: CSMA = pehle suno, khali ho toh bhejo

2. **Collision:**

When two computers transmit simultaneously, their signals overlap — this is a collision. Data from both is garbled (but hardware is not damaged).

3. **Ethernet CD (Collision Detection — CSMA/CD):**

Ethernet hardware monitors outgoing signals. A garbled signal means a collision is detected. Both computers stop transmitting.

4. **Recovery from Collision:**

a. Computer sends a jam signal to force all to detect the collision

b. Both computers wait a random time before trying again

c. Computer with shorter delay goes first

5. **Exponential Backoff:**

With each repeated collision, the wait time doubles. This reduces the chance of repeated collisions on a busy network.

Hint: Collision ho = random wait, baar baar ho = wait double karo